OAK RIDGE National Laboratory

AI-Agent Driven Survey and Validated Literature Review for Discovering Solid Breeder Materials

End-to-End Traceability for Automated Literature Generation

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Solid Breeders are a Bottleneck for Fusion Energy

Current Approaches

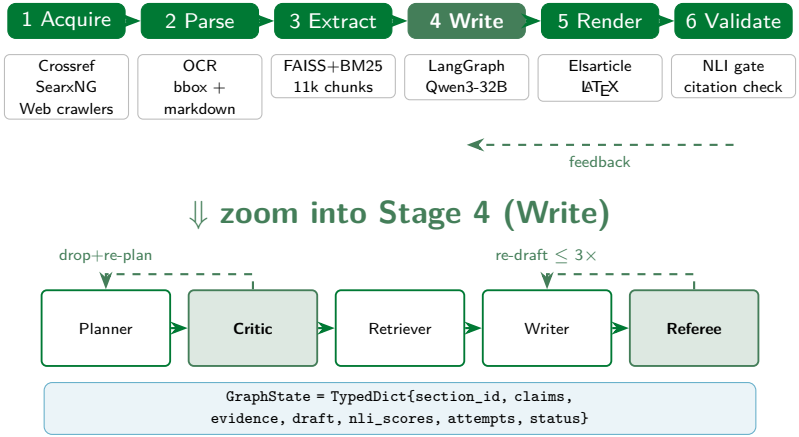
- Analyze papers and databases individually
- Limited claim-level verification
- Static LLM prompts cannot explore research questions in depth
- Focus on prediction, not review generation

Our Approach

- Integrate multimodal information (incl. ontologies)
- Verifies every sentence
- Produces traceable review manuscripts automatically
- Graph-based approach enables deeper exploration of research questions

References: Zhou *et al.*, *Int. Dent. J.* 2026 doi:10.1016/j.identj.2025.07.013 ○ Tasdemir, *J. Alloys Metall. Syst.* 2026 doi:10.1016/j.jalmes.2025.100192 ○ Tang *et al.* (MatterChat), *Nat. Mach. Intell.* 2026 doi:10.1038/s42256-025-01065-z ○ Zhang *et al.*, *Adv. Funct. Mater.* 2026 doi:10.1002/adfm.202508472.

Six Stages End-to-End | Writing Stage Expands Into a 5-Agent LangGraph With Shared State



Why a graph: Breaks broad scientific questions into focused sub-questions; shared state records sources and decisions for per-claim traceability.